



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

B. Tech CSE

Course Name: Introduction to Full Stack Development

Scheme [L-T-P]: [3-0-0]

Course Learning Outcomes:

1. Understand core web technologies such as HTML, CSS, JavaScript, and responsive design.
2. Build interactive front-end applications using modern JavaScript frameworks like React.
3. Develop and test scalable backend applications using Node.js and Express with NoSQL databases like MongoDB.
4. Apply full-stack development principles to create complete web applications

Sr.No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Frontend Web Development - HTML5 elements and attributes - CSS3 styling, box model, layout techniques - Responsive design with Bootstrap - Interactive media and form handling	10	10
2	JavaScript Programming & DOM Manipulation - JavaScript syntax, functions, and ES6 features - DOM manipulation and event handling - Building interactive components - Mini projects: QR code generator, weather app	8	8
3	ReactJS Fundamentals - JSX and Virtual DOM	8	8



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	- Components, props, and state - Forms and HTTP requests - Routing with React Router		
4	Backend Development with Node.js - Architecture and asynchronous programming - Creating REST APIs using Express.js - Middleware, routing, and error handling - Unit testing with Jest	8	8
5	Database Integration with MongoDB - Introduction to NoSQL databases - CRUD operations, collections, and documents - MongoDB client, aggregation, and indexing, - Integration with Node.js	6	6
6	Full Stack Project Implementation - Project planning and structure - Frontend-backend integration - Deployment overview (Heroku/Vercel) - Project demo and presentation	5	5

Textbooks and References

1. E. Freeman and E. Robson, 'Head First HTML and CSS', O'Reilly Media.
2. Jon Duckett, 'JavaScript and jQuery: Interactive Front-End Web Development', Wiley.
3. Stoyan Stefanov, 'Learning JavaScript Design Patterns', O'Reilly Media.
4. Alex Banks and Eve Porcello, 'Learning React', O'Reilly Media.
5. Brad Dayley and Brendan Dayley, 'Node.js, MongoDB and Angular Web Development', Addison-Wesley